

Empirical Profiling of Cloud-Based Healthcare Interoperability Gateways

Master's Thesis Presentation

Alberto Angeloni

University of Padua

Academic Year 2025–2026

25 September 2026



UNIVERSITÀ
DEGLI STUDI
DI PADOVA



INFORMATICA
DEPARTMENT OF INFORMATICS

Narrative Path

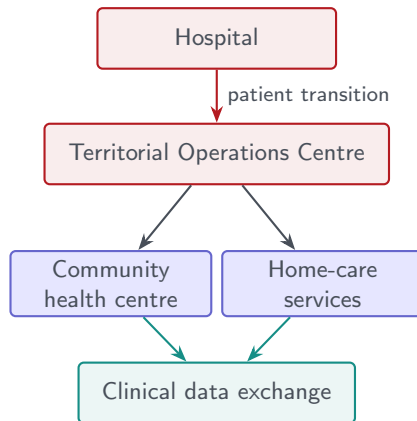


From healthcare needs to defensible conclusions

Territorial Care Depends on Interoperability

01 / 05

- **Policy** Ministerial Decree 77/2022 reorganises territorial care
- **Transitions** One patient journey spans hospital community and home care
- **Coordination** The Territorial Operations Centre connects care settings
- **Dependency** Care continuity requires timely clinical data exchange



Care transitions depend on a shared information path

Thesis Question and Evaluation Lenses

01 / 05

PROBLEM

Endpoint tests alone do not preserve the semantics of an ordered healthcare workflow

THESIS QUESTION

WORKLOADS

How can documented workflows become reproducible end-to-end workloads

ESTIMATION

How can apparent mediation overhead be estimated without gateway telemetry

FOUR EVALUATION LENSES

Semantics

Ordered dependencies and concurrent state

Reproducibility

Flow mix arrivals identifiers artefacts

Evidence integrity

Request flow and window populations

Behaviour

Latency reliability completion under load

The lenses operationalise the single thesis question

Declared Paths Share One Observation Boundary

02 / 05

INPUT

Documented regional workflows

CONTROL

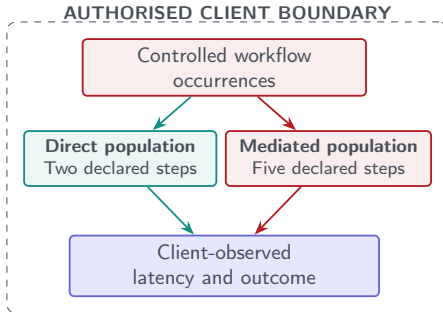
Executable occurrences with configured flow mix and arrivals

COMPARISON

Declared direct and mediated populations

MEASUREMENT

Client send-to-response timing and terminal outcomes



The comparison is path-level rather than gateway-only

Workflows Produce Traceable Evidence

02 / 05

REPRODUCIBILITY

CRITERION

Stable identifiers link configuration workload events and outcomes

OBSERVED

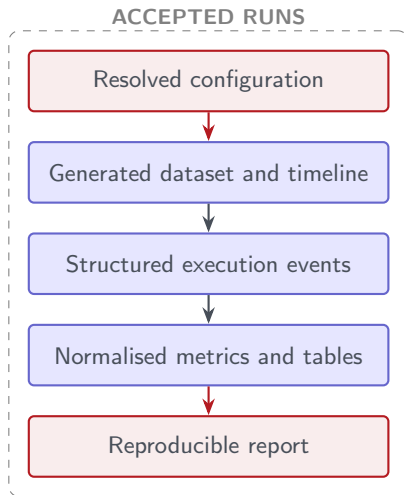
All 74 accepted runs retain the complete artefact chain

CONSISTENCY CHECK

$757,517 = 2 \times 299,534 + 158,449$

BOUNDARY

Gateway-internal fields remain empty by design



The Realised Evidence Is Smaller Than Planned

02 / 05

PLANNED

300 campaign runs

STARTED

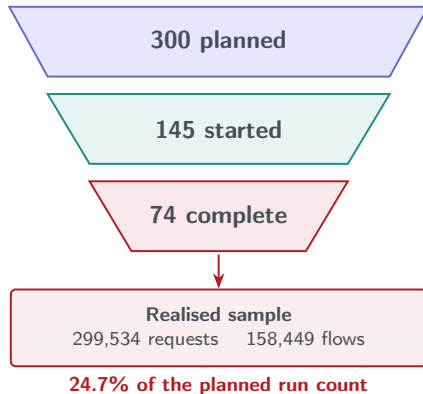
145 runs with execution evidence

COMPLETE

74 accepted runs across ten scenarios

LIMIT

Incomplete protocol and unbalanced repetitions support descriptive rather than population-wide claims



Successful-Flow Median Displacement

03 / 05

EXPECTED

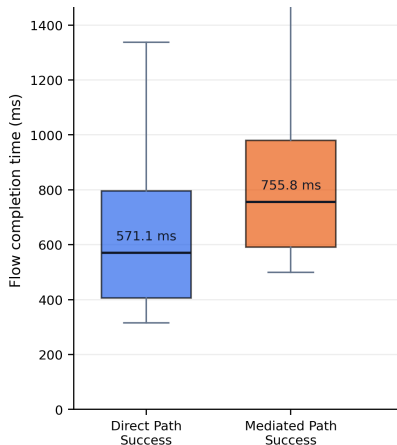
Mediated P50 exceeds direct P50 for successful flows

OBSERVED

571.1 ms direct and 755.8 ms mediated with $\Delta P_{50} = +184.7$ ms

INTERPRETATION

Positive client-observed path displacement



Empirical successful-flow distributions at the client boundary

Early Failure Truncates the Flow Prefix

03 / 05

EXPECTED

An unsuccessful transition suppresses its dependent suffix

OBSERVED

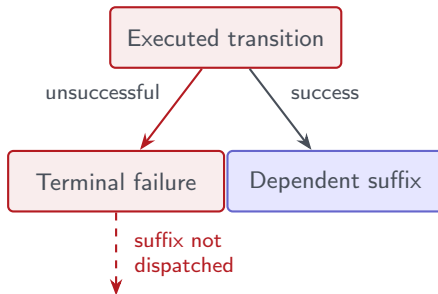
+12.88 ms failed versus
+184.7 ms successful

INTERPRETATION

The smaller gap is consistent with a shorter realised prefix

LIMIT

12.88 ms is a path differential rather than a failure duration



Only executed transitions contribute to the realised prefix

Admission Saturation Precedes Delay Growth

03 / 05

DESIGN BOUND

Active client flows $\leq C_{\max} = 30$

OBSERVED

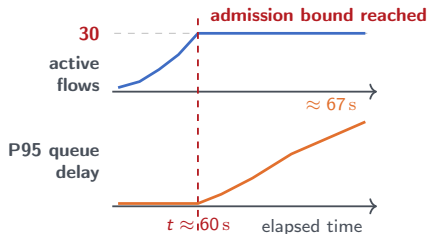
At $t \approx 60$ s active flows reach 30 while P95 queue delay later approaches 67 s

INTERPRETATION

Queued occurrences create client-side backpressure

LIMIT

The plateau does not measure regional gateway capacity



Extreme Load Deforms the Latency Distribution

03 / 05

EXPECTED

Successful-flow distribution under extreme versus baseline load

OBSERVED

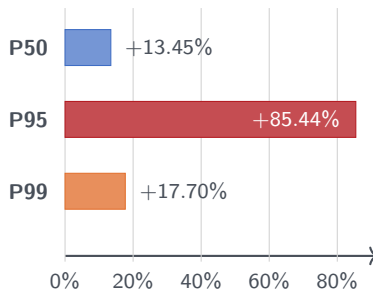
P50 +13.45% P95 +85.44%
P99 +17.70%

COMPLETION

Flow success decreases by 3.63 percentage points

INTERPRETATION

The upper-tail shoulder changes most



Flow success -3.63 percentage points

Extreme stress relative to the baseline constant scenario

Supported Claims and Evidential Boundaries

04 / 05

SUPPORTED

PATH CHARACTERISATION

Client-observed distributions and path-level differentials

WORKFLOW BEHAVIOUR

Fail-fast suffix suppression and terminal outcomes

INSTRUMENT LIMIT

Client backpressure after the configured bound

NOT ESTABLISHED

INTERNAL CAUSALITY

Gateway-only processing time

REMOTE CAPACITY

Provider capacity or a service-level guarantee

GENERALISATION

Production-wide traffic representativeness

74 complete runs unbalanced repetitions unequal path composition

The Thesis Delivers a Bounded Empirical Profile

04 / 05

EXECUTABLE MODEL

Ordered healthcare workflows
become controlled occurrences

EVIDENCE MODEL

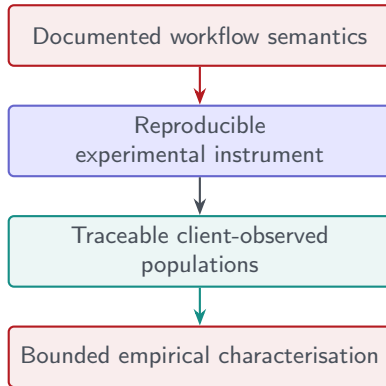
Identifiers join configuration events
requests flows and reports

EMPIRICAL PROFILE

Path displacement fail-fast
behaviour and load-sensitive tails
become observable

CLAIM DISCIPLINE

Every conclusion remains inside
the realised client-observed
perimeter



The contribution is the complete chain rather
than one isolated metric

Answer to the Thesis Question

05 / 05

Reproducible workloads

Documented dependencies become traceable occurrences across 74 accepted live runs



Apparent mediation overhead

The declared successful-flow populations differ by +184.7 ms at P50 within the client boundary



Evidential boundary

Client saturation and unequal path composition prevent gateway-only or capacity claims

TAKEAWAY

RDG makes protected healthcare workflows experimentally observable without claiming visibility it does not have